

Greater Survival After Breast Cancer in Physically Active Women With High Vegetable-Fruit Intake Regardless of Obesity

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Purpose Single-variable analyses have associated physical activity, diet, and obesity with survival after breast cancer. This report investigates interactions among these variables. **Patients and Methods** A prospective study was performed of 1,490 women diagnosed and treated for early-stage breast cancer between 1991 and 2000. Enrollment was an average of 2 years postdiagnosis. Only seven women were lost to follow-up through December 2005. **Results** In univariate analysis, reduced mortality was weakly associated with higher vegetable-fruit consumption, increased physical activity, and a body mass index that was neither low weight nor obese. In a multivariate Cox model, only the combination of consuming five or more daily servings of vegetables-fruits, and accumulating 540+ metabolic equivalent tasks-min/wk (equivalent to walking 30 minutes 6 d/wk), was associated with a significant survival advantage (hazard ratio, 0.56; 95% CI, 0.31 to 0.98). The approximate 50% reduction in risk associated with these healthy lifestyle behaviors was observed in both obese and nonobese women, although fewer obese women were physically active with a healthy dietary pattern (16% v 30%). Among those who adhered to this healthy lifestyle, there was no apparent effect of obesity on survival. The effect was stronger in women who had hormone receptor-positive cancers. **Conclusion** A minority of breast cancer survivors follow a healthy lifestyle that includes both recommended intakes of vegetables-fruits and moderate levels of physical activity. The strong protective effect observed suggests a need for additional investigation of the effect of the combined influence of diet and physical activity on breast cancer survival.